

2-HOUR CRAM — Computer Apps / ICT in Agriculture

Source PDF: agri input automation → models → apps → GIS/RS/GPS → DSS → expert systems

Exam map: SVU-style 2nd year (1 / 3 / 6 mark tags).

How to use (120 min): 0–15 mark legend + unit map · 15–70 units A–C · 70–100 units D–F · 100–120 write 2× 6-mark skeletons + flash gotchas.

Mark legend

Tag	Meaning	Write length
[1]	Definition / one fact / expand abbreviation	1–2 lines
[3]	Short note / list + one line each	½ page
[6]	Full answer: def → parts → agri use → limit/example	1–1½ pages

0. One map of the whole syllabus

SENSE the field → MODEL what will grow → ADVISE the farmer → ACT on map
sensors/GPS crop simulation apps / expert / DSS VRA / irrigation
↑
GIS holds layers

Memory hook: **S-M-A-A** — Sense · Model · Advise · Act.

A. Computer-controlled devices (agri input management) [3–6]

First principle

A computer-controlled farm device **measures** (sensor) → **locates** (GPS/geospatial) → **decides rate** (controller) → **applies input** (seed/fertilizer/water).
Precision farming = right amount, right place, right time.

Five systems (list for [3] / expand any one for [6])

Device	What it does	Exam one-liner
Sensor	Converts heat/light/pressure into a computer signal	Barcode scanner = light → code (needs line of sight)
Seed planting system	Shoes plant seed; sensors log each drop	Optimal spacing under computer control
Geospatial locator	Sends lat/long of each seed to computer	Builds seed distribution profile
Variable Rate Application (VRA)	Rate of material depends on location / field quality	Not one flat rate for whole field
Irrigation system	Water by need / map / schedule under control	Links to sensors + VRA idea

Mnemonic for five devices: **S-S-G-V-I** — Sensor, Seed, Geospatial, VRA, Irrigation.

Extra PDF facts (do not skip in long answers)

- Sensors also fit **greenhouses / polyhouses** (protected cultivation) and precision farming.
- Barcode: optical read by light. Needs **line of sight** (no obstruction).
- Daily-life computer control example in notes: **ATM** (balance, deposit, withdraw).
- **Robots** = programmed move / manipulate / interact with environment (opening example).
- Seed system: mobile planter with **planting shoes** (towed or self-propelled). Sensors near shoes log plant events to a computer.
- VRA kit examples: computer/controllers, liquid sprayers, **granular fertilizer** applicators, air sprayers/spreaders, drills and planters.

Gotchas

- Sensor is **not** the same as GIS. Sensor = data capture. GIS = store/analyze maps.
- Barcode needs **line of sight** (obstruction fails).
- VRA needs **location quality data** (soil/yield map). Without map, VRA is just a sprayer.

Expected Qs

1. [1] Define sensor.
2. [3] Write short notes on Variable Rate Application.
3. [6] Explain computer-controlled devices for agri input management with examples.

B. Computer / crop models [3–6]

First principle

Model = simplified picture of a system in math form.

Simulation = run that picture through time with weather/soil/management to forecast growth and yield.

Why models [3] (list 4)

1. Store knowledge from many field trials in one structure.
2. Force multi-discipline teamwork.
3. Use systems analysis on complex crops.
4. Give dynamic, quantitative tools for management choice.

Two types of crop growth models [1–3]

Type	Idea	Exam word
Regression / empirical	Fit a curve (Richards, polynomials) to growth data	Describes pattern; weak on “why”

Mechanistic	Build from physiology (photosynthesis, water, N) vs environment	Explains mechanism
--------------------	---	--------------------

Inputs you must list [3–6]

Class	Key items
Weather (daily)	Tmax, Tmin, rain, RH, solar radiation, wind
Crop	Name, variety, phenology (days to anthesis/maturity), LAI, biomass, 1000-grain weight
Soil	Layer thickness, pH, EC, NPK, SOC, texture, sand/clay %, moisture, saturation, FC, WP, bulk density
Management	Sowing date (or transplant date for rice), seed rate/depth, irrigation, fertilizer, manure, residue (type, date, depth). C:N if organic
Pests	Insect-pest data can enter the modeling package when the syllabus lists it

Start time gotcha: Simulation usually starts at **sowing**. For transplanted rice, use **transplant date**, not nursery sowing.

Model definition line for [1]: A model is a simplified representation of a system. Modelling writes the process as formal math. Simulation builds the model and analyzes the system over time.

Uses [3]

Pre-season / in-season decisions: fertilizer, irrigation, pesticides. Policy: erosion, leaching, climate effect, large-area yield forecast.

Limits (add in [6] for full marks)

You need good daily weather. Wrong variety parameters give wrong yield. Simple models are weak on pests. Bad inputs give bad output.

Expected Qs

1. **[1]** Define crop model / simulation.
2. **[3]** Advantages of crop simulation models.
3. **[3]** Differentiate empirical vs mechanistic models.
4. **[6]** Components and input data requirements of crop simulation models.

Hack: Write **W-C-S-M** for inputs: Weather, Crop, Soil, Management.

C. Smartphone apps (advice · market · postharvest) [3–6]

First principle

App = software on phone/tablet (not desktop/laptop). Moves **weather, price, scheme, expert chat** to the field so the farmer decides faster. Apps can ship **preloaded** or **download** from an app store / internet.

Why phones help: cloud computing + integrated IT + online education + mobile reach poorest communities. Better land decisions (soil + weather). Optimize fertilizer, seed, water. Save money and cut waste.

India note in PDF: apps from **Jayalaxmiagrotech** called among the most used agri apps in India. Domains: agriculture, horticulture, animal husbandry, farm machinery.

App types [1]

Type	Meaning
Native	Built for one OS; uses device features fully
Hybrid	Mix of native + web
Web app	Stored on remote server. Delivered in a browser

Farm-advice apps (name 5 for [3])

App	Remember for
AgriApp	Marketplace + expert chat + videos (~0.1M users, ~4.3★)
IFFCO Kisan	Weather, mandi rates, advisories, ~11 Indian languages, expert queries
Agri Media Video	Videos + chat + upload infected crop image (~4.8★)
Farm-Bee / RML Farmer	~450 crop varieties, ~1300 markets, ~3500 weather points; multi-language; ~0.5M users
KisanYojana	Govt schemes gap-closer (saves trip to office)
Smart Krishi	Package of practices (high value low volume crops), success/fail stories, Krishi library, GPS farm contacts, tips, live weather (DHM/MFD)

RML: Reuters Market Light — subscription **SMS** for local **commodity price, crop cultivation tips, weather**.

IFFCO Kisan feature family (list if asked): agro IT · call centre · urban grading · software solutions · commodity services · rural distribution.

Market price apps [3]

App	One fact
AgriMarket	Prices within 50 km via GPS (or manual market pick)
eNAM	Pan-India electronic trading; links APMC mandis ; SFAC implements under MoA&FW
Rythu Bazar App	Local vegetable market prices + nearest Rythu Bazar
Mandi app	Live mandi rates; zone-wise APMC; daily sync

eNAM must-know [1–3] (high frequency)

- **What:** Unified national electronic market for agri commodities. Networks existing **APMC mandis**.
- **Who:** **SFAC** is the lead agency under the Ministry of Agriculture and Farmers' Welfare.
- **Why:** Real-time price discovery. Cut information asymmetry buyer–seller. Support online payment.
- **Vision:** Uniform marketing procedures. Remove information asymmetry. Real-time price discovery from demand and supply.
- **Mission:** Integrate APMCs on one online platform. Transparent quality-based auction. Timely online payment.

- **Stats often asked: 90+ commodities.** Mobile app in **8 languages**. Trade mostly intra-market first. Then inter-market / inter-state phases.

AGMARKNET: Agricultural Marketing Information Network. Launched **March 2000** by Union Ministry of Agriculture (pair with mandi price apps).

Postharvest golden rule [1] (almost always asked)

Quality cannot be improved after harvest — only maintained.

So harvest only good produce. Then cool, clean, sort, and pack. Poor quality means short postharvest life. Cultural practices **before** harvest still matter for market life.

CHEETAH app (Africa example): Chains of Human Intelligence for efficiency/equity in agro-food trade along the Trans-Africa Highway. Targets losses when farmers cross borders.

Loss numbers in PDF (1-mark bait): Africa postharvest food losses about **\$48 billion/year**. About **33%** of global food production lost. Example: bruised lots sorted out → farmer can lose about **30%**.

Expected Qs

1. **[1]** Postharvest golden rule.
2. **[1]** Expand eNAM / SFAC.
3. **[3]** Types of mobile apps with examples.
4. **[3]** Any four agriculture mobile apps and features.
5. **[6]** Role of smartphone apps in farm advice and market price discovery.

Hack: A-I-F-K-E — AgriApp, IFFCO, FarmBee, KisanYojana, eNAM.

D. Geospatial technology: GIS · RS · GPS [6 heavy]

First principles (write this first in any long answer)

Term	Plain meaning
GIS	Computer system for geo-referenced data: capture, store, edit, analyze, display
Spatial data	Where (point, line, polygon, surface)
Attribute data	What (name, yield, soil class) — non-location properties
Remote sensing	Measure object/area properties without physical contact (acoustic or EM; agri mostly electromagnetic spectrum)
GPS	Satellite positioning for field coordinates
Data model	Abstract description of how data and relationships are stored in the system

ESRI-style definition (use in **[1]/[6]**): organized hardware + software + geographic data + people to capture, store, update, manipulate, analyze, and display geographically referenced information.

RS on farm [3]: Eye-check of leaf color without touch is primitive RS. Satellite/aircraft images assess nutrient lack, disease, water stress, insect/hail/wind/herbicide damage, plant population. Feed **base maps** for VRA of fertilizer and pesticide. Treat only affected zones.

GIS components [3] (two answer styles — both valid)

Style A (process): 1) Data input 2) Storage & retrieval 3) Analysis & manipulation 4) Output

Style B (system): Hardware · Software · Data · People (skilled users)

Software module group: input/verify · storage/DB · transform · output · user interaction.

Raster vs vector [3] (classic 1–3 mark)

	Raster	Vector
Form	Grid of cells/pixels	Points, lines, polygons (+ TIN for surfaces)
Best for	Continuous fields (elevation, temp, NDVI)	Discrete objects (roads, fields, wells)
Trade	Simple math; can be heavy on storage	Precise boundaries; topology

Conceptual models: **Objects** (discrete boundaries) vs **Fields** (value everywhere — elevation, soil chemistry).

Logical models: **Raster** (grid) · **Vector** (points/lines/polygons; continuous fields can use **TIN**) · **Object-oriented** (real-world objects as base, not only pure geometry).

Four feature types to store: point (0-D) · line (1-D) · area/polygon (2-D) · continuous surface.

Five Ms of GIS application [1–3]

Mapping · Measurement · Monitoring · Modeling · Management

GIS application domains [3]

Government/public service · business and service planning (**geodemographics** for market areas) · logistics and transportation · environment (sprawl, urban impact).

GIS analysis ops (name any 5 in [3])

Retrieval · buffer · overlay/dissolve · measurements · network · digital terrain analysis · map generalization/transform/export.

GIS benefits (3 lines for [3])

Integrate many databases · spatial analysis in complex settings · fast specialized maps · manage data in spatial context.

Popular packages [1]

ArcGIS, AutoCAD, Cadcorp, ERDAS IMAGINE, IDRISI, Intergraph, MapInfo, MicroStation (open-source options also exist).

Geodatabase [1]: One container for spatial + non-spatial data. Consolidated storage. Can apply **validation rules** on attributes.

Remote sensing in farm [3]

- Farmer eyeing leaf color without touching plant = primitive RS.
- Modern: digital imagery + GIS for crop stress, management zones.
- Pairs with GPS waypoints and yield maps.

GPS + GIS in agriculture (precision farming) [6]

Precision / site-specific farming needs **GPS + GIS** (PDF typo “CPS” = **GPS**).

Use	What happens
Tractor guidance	Record path; repeat for cultivate / fertilize / harvest
Crop-duster targeting	Mark pest hotspots; spray only those zones
Livestock tracking	Collar transmitters. Track animals to market.
Soil sampling	Waypoint samples. Plot lab results on the map. Treat only needy zones.
Yield monitoring	Zone yields on a map. Plan the next season.

Also: GPS maps **field boundaries**, **roads**, **irrigation systems**, and problem spots (weeds, disease). Saves money when you treat only documented need.

Hack: T-C-L-S-Y — Tractor, Crop-duster, Livestock, Soil sample, Yield.

Gotchas

- GIS is **not** only maps. Analysis and query are the exam point.
- Attribute without location is a normal database. GIS **requires** a spatial link.
- Raster is not photo only. Any grid (soil grid, NDVI) is raster.
- The source PDF can print “CPS”. Write **GPS** in the answer.

Expected Qs

1. **[1]** Define GIS / remote sensing / spatial data.
2. **[1]** Five Ms of GIS.
3. **[3]** Components of GIS.
4. **[3]** Raster vs vector.
5. **[6]** Applications of GIS and GPS in agriculture / precision farming.
6. **[6]** Concept, components and uses of geospatial technology in DSS context.

E. Decision Support System (DSS) [3–6]

First principle

DSS = interactive computer system that helps a person use **data + models + knowledge + communication** to **structure a decision**. It does not replace the decision maker.

Three core architecture parts [1–3]

1. **Database / knowledge base**
2. **Model** (decision context + criteria)
3. **User interface**

(+ user as part of the system)

DSS types [3] (memorize order)

Type	Emphasizes	PDF example
Model-driven	Optimization / simulation; limited user parameters	Sprinter, MEDIAC, Brandaid
Data-driven	Time-series internal/external data; queries	WalMart multi-terabyte scale
Communication-driven	Groupware, video conference, bulletin boards	Collaboration tools
Document-driven	Policies, catalogs, scanned/historical docs	Retrieval and analysis
Knowledge-driven	Recommend actions (AI / expert systems)	Scheduling, web advisory

Mnemonic: M-D-C-D-K — Model, Data, Comm, Document, Knowledge.

Decision phases (Early framework) [1–3]

Intelligence → **Design** → **Choice** → **Implementation**

(search problem → invent options → pick → adopt)

Tools under DSS umbrella [1]

OLAP / multi-dimensional analysis · Query tools · Data mining · also GIS, EIS, Expert Systems, software agents, knowledge discovery, group DSS.

DSS technology levels [1–3]

1. **Application** — what the decision maker uses on a problem.
2. **Generator** — build environment (Crystal, Analytica, iThink).
3. **Tools** — languages, libraries, link modules.

Other classifications (if paper asks “classify DSS”) [3]

- **Holsapple & Winston (6)**: text · database · spreadsheet · solver · rule · **compound** (hybrid; most popular).
- **Support level**: Personal · Group · Organizational.
- **IO view**: Inputs · user knowledge · outputs · decisions.
- **IDSS**: intelligent DSS (AI / agents).
- Non-agri PDF examples: clinical DSS (CDSS); Canadian National Railway derailment/wear testing.

When to build [3] (checklist style)

Can conventional code solve it? Domain well-bounded? Need/desire for expert system? Cooperating human expert? Expert can explain knowledge? Knowledge mainly heuristic/uncertain?

Agri example [1–3]

DSSAT4 (USAID-era support): rapid assessment of production systems for farm and policy decisions. Precision agriculture = decisions tailored to **portions of fields**. Adoption still has constraints. Forest management is another major DSS domain.

Expected Qs

1. **[1]** Define DSS.
 2. **[3]** Components of DSS.
 3. **[3]** Types of DSS with examples.
 4. **[6]** DSS in agriculture and precision farming / DSSAT.
-

F. Agriculture expert systems & Soil Information Systems [3–6]

First principle

Expert system = program that solves problems by **mimicking expert reasoning** (rules, experience, logic) — not only number-crunching formulas. Needed because farm problems are often **qualitative** and multi-discipline.

Expert system building blocks [3]

User interface · Database · **Knowledge base** · **Inference mechanism**

Dev phases [3]: problem selection → knowledge acquisition → knowledge representation → programming → testing and evaluation.

Why farming needs them [3]

Yield loss, erosion, crop choice, pesticide cost/resistance, price pressure, strategy barriers — one farmer cannot be expert in pathology + entomology + meteorology at once. ES integrates those into day-to-day advice.

Soil Information System (SIS) [3]

High-resolution soil + topography maps via sensors + geo-processing. Characterizes physical/chemical soil and how inputs move. Supports irrigation, drainage, fertility by **zone**, not average field.

Benefits (farmers) [3]

Better decisions when specialist absent · higher profit via input save · sustainability (less runoff) · training tool for new staff.

Name 4 web ES + 4 mobile [3]

Web / portals: Maize Agri Daksh (IASRI) · Wheat ES (IASRI) · Digital mandi for Indian kisan · mKisan Agri portal · RICE Doctor (IRRI) · TNAU AgriTECH PORTAL · Barley expert system · Rice Knowledge Management Portal · ES agri & animal husbandry (DWCRA, Bhubaneswar)

Mobile: Crop insurance (MoA, GoI) · AgriMarket · mKisan · RainbowAgri · Manditrades · Mpower social · IFFCO Kisan · eSAP

Expected Qs

1. **[1]** Define expert system / SIS.
 2. **[3]** Components of an expert system.
 3. **[3]** Importance of expert systems in agriculture.
 4. **[6]** Expert systems and soil information systems for farm decisions (with examples).
-

G. Last-hour answer skeletons (copy structure)

Skeleton [6] — GIS and GPS in agriculture

1. Def GIS + GPS (3 lines).
2. Precision farming link (1 line).
3. Five uses T-C-L-S-Y with one sentence each.
4. One limit (cost/skill/data quality) + conclusion.

Skeleton [6] — Crop simulation models

1. Def model + simulation.
2. Why needed (4 bullets).

3. Empirical vs mechanistic (table).
4. Inputs W-C-S-M (+ rice transplant gotcha).
5. Uses + one limitation.

Skeleton [6] — Computer-controlled devices / VRA stack

1. Precision idea.
2. Sensor → seed system → geospatial → VRA → irrigation.
3. One worked flow: soil map + GPS → VRA fertilizer.
4. Limit: bad map → bad rate.

Skeleton [6] — DSS / Expert system

1. Define DSS vs ES. DSS is a wider toolkit. ES uses knowledge base plus inference.
2. Architecture parts.
3. Types or components.
4. Agri examples (DSSAT, IFFCO or mKisan class tools, SIS).
5. Close with decision quality and sustainability.

H. Rapid flash — 40-second drills

Cue	Answer
PHM golden rule	Quality not improved after harvest — only maintained
eNAM implementer	SFAC under MoA&FW
AgriMarket radius	50 km GPS
Five Ms	Map Measure Monitor Model Manage
DSS core 3	DB · Model · UI
DSS phases	Intelligence Design Choice Implementation
Model types	Empirical vs Mechanistic
Rice model start	Transplant date
Raster	Grid cells
Vector	Point line polygon
ES core	Knowledge base + inference
VRA	Rate by location/quality
Sensor	Stimulus → distinctive signal
RML	Reuters Market Light SMS

I. Exam gotchas (lose marks if you mix these)

1. **GIS vs GPS vs RS** — GPS locates. RS senses at a distance. GIS analyzes layers.
2. **Model vs simulation** — model is structure. Simulation is the run of that structure.
3. **Native vs web app** — native = platform-specific install. Web = browser.
4. **Attribute ≠ spatial** — GIS needs both.
5. **eNAM ≠ Rythu Bazar** — eNAM is the national electronic APMC network. Rythu Bazar is a local fresh market app.
6. **Expert system ≠ spreadsheet** — ES uses inference and knowledge. It is not only formulas.
7. Spelling traps: **geospatial, phenology, inference, APMC, SFAC**.

J. 15-minute pre-exam order

1. Write PHM rule + eNAM + Five Ms + raster/vector from memory.
2. Sketch S-M-A-A map once.
3. Recite T-C-L-S-Y and W-C-S-M.
4. Open answer with definition + one agri example — examiners scan first 5 lines.

Source scope: Your provided PDF unit on computer-controlled inputs, models, apps, GIS/RS/GPS, DSS, expert systems & SIS. Align wording with class handouts if teacher used different lists.

K. Core questions (answered in body)

1. What is the farm ICT chain? → S-M-A-A (Sense Model Advise Act).
2. What is VRA? → Rate of input by location or field quality.
3. What data does a crop model need? → Weather, Crop, Soil, Management (W-C-S-M).
4. What is the postharvest rule? → Quality cannot improve after harvest. Only maintain it.
5. What is GIS vs GPS vs RS? → Analyze layers vs locate vs sense without contact.
6. What is a DSS? → Interactive aid for decisions with data, models, and interface.
7. What is an expert system? → Program that mimics expert inference with a knowledge base.